

A Hayekian Flatcoin: Commodity-Reserve Logic for a Blockchain Monetary Standard

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Abstract: This paper proposes a Hayekian flatcoin, i.e. a blockchain-based currency whose value is defined by a continuously adjusted basket of commodities and maintained through competitive private issuance. It begins from the observation that contemporary stablecoins reproduce the inflationary bias of state-issued money by pegging to fiat currencies. Much of the cryptocurrency literature draws on F. A. Hayek's 1970s writings, notably *Choice in Currency* and *Denationalisation of Money*, to justify this practice, yet it neglects his 1943 essay *A Commodity Reserve Currency*, which offers a concrete mechanism for a money of stable purchasing power. Revisiting that earlier proposal and drawing on later Austrian monetary theory and historical evidence of commodity standards, the paper shows how modern blockchain technology can implement Hayek's rule-bound design. The result is a monetary standard that fulfils Hayek's aim of depriving governments of the monopoly of money issue and Satoshi Nakamoto's ambition for peer-to-peer electronic cash. A Hayekian flatcoin would offer a credible, inflation-resistant alternative to fiat-backed stablecoins and illustrate how entrepreneurial innovation can extend the market order in the digital age.

Keywords: Commodity reserve currency; Cryptocurrency; blockchain; decentralised autonomous organisation.

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Introduction

Stablecoins now dominate the monetary layer of blockchain applications. They have enabled large-scale decentralised finance and global transfers, yet their architecture remains tied to the monetary order that the blockchain revolution set out to escape. The leading examples are liabilities of private issuers that hold dollar-denominated assets in conventional banks and money-market instruments. They are, in effect, private representations of central bank money. This structure imports the discretionary inflation and political risk of the underlying fiat system and depends on legal enforcement by the state. It falls short of Satoshi Nakamoto's aim, set out in the Bitcoin white paper, to create 'a purely peer-to-peer version of electronic cash' that allows payments to be sent 'without going through a financial institution' (Nakamoto, 2008, p.1).

This paper recovers a neglected strand of F. A. Hayek's thought. In 1943 he outlined a commodity-reserve currency designed to preserve purchasing power through a market-based adjustment of a basket of goods (Hayek, 1943). Later writings, especially his 1970s pamphlets on currency competition (Hayek, 1976; 1976/78), have been more influential in cryptocurrency debates, yet they left the unit of account tied to state money. By revisiting the 1943 proposal, the analysis shifts attention from competition in issuance to the design of money itself.

Re-examining Hayek's 1943 argument speaks directly to contemporary problems. Empirical evidence shows that commodity standards have historically delivered lower and more predictable inflation than fiat standards (Rolnick and Weber, 1997; White, 2015). The theory of competitive supply of private money is well established (Klein, 1974; Selgin, 1988). Bitcoin and related cryptocurrencies demonstrate that decentralised monetary systems can gain global acceptance without legal tender status (Hendrickson, Hogan and Luther, 2016; Luther, 2019). Recent research on blockchain and decentralised autonomous organisations shows how verifiable rules and credible exit can substitute for administrative control (Davidson, 2023; 2025a; 2025d). Together these insights suggest that a Hayekian flatcoin – an inflation-resistant token anchored to a commodity basket and governed by decentralised rules – can now be designed and issued.

The argument is developed within the Austrian tradition that treats markets as processes of discovery and institutions as the rules of the game (Hayek, 1945; Selgin and White, 1994). It combines historical monetary theory with the institutional possibilities of blockchain. Section 2 examines why present stablecoins, by remaining tied to fiat, fall short of Satoshi's ambition. Section 3 reconstructs the commodity reserve currency and places it in the context of its 1943 exchange with John Maynard Keynes, explaining how a basket of goods and competitive private issuance can deliver stable purchasing power without central bank management. Section 4 translates this design into an on-chain flatcoin, explaining the roles of blockchain, smart contracts, oracles and decentralised autonomous organisations (DAOs). Section 5 considers the economic and institutional implications of such a system, including issues of volatility, network effects, legal treatment and the potential for 'repugnant innovation' (Allen, Berg and Davidson, 2023). A final section concludes by identifying the conditions under which a Hayekian flatcoin can emerge as a non-state digital money and by outlining its significance for both Austrian monetary theory and contemporary cryptoeconomics. For those readers unfamiliar with blockchain terminology, an appendix provides a glossary of terms.

Fiat Anchoring and a Misreading of Hayek

The most widely used stablecoins, such as USDC and USDT, are promises to redeem on demand for United States dollars held in bank accounts or money-market instruments. They provide on-chain liquidity but do not create a new monetary order. They are a private settlement layer built on top of the existing fiat system. Their stability depends on the discretion of central banks and on the solvency

and honesty of political actors. In this respect they replicate the very conditions that Satoshi Nakamoto's design for Bitcoin was intended to overcome (Nakamoto, 2008). Fiat-backed stablecoins fail this test because their value and redemption ultimately rest on those very same financial institutions.

This arrangement reflects a selective reading of Hayek. In *Choice in Currency* Hayek urged that individuals should be free to choose among competing monies and that 'governments should be deprived of their monopoly of issuing money' (Hayek, 1976, p.22). He repeated this point in *Denationalisation of Money*, arguing that 'the abolition of the government monopoly of issuing money would indeed be the quickest and most certain way to stop inflation' (Hayek, 1976/1978, p.14). Written during the inflation of the 1970s, these pamphlets have become canonical in cryptocurrency debates about private money and are widely cited to justify the issue of dollar-denominated tokens on the ground that competition disciplines issuers.

Yet a focus on competition in note issue is not the same as a new standard of value. Hayek's later writings assumed that national currencies would remain the unit of account and concentrated on opening the supply of that unit to rival issuers. That approach leaves open, and even encourages, the continuation of private liabilities denominated in fiat money. Most stablecoins do precisely this. They are competitive in the limited sense of allowing more issuers, but they remain dependent on the monetary policy and political risks of the state.

Hayek's earlier work points to a different possibility. In *A Commodity Reserve Currency* he outlined how a privately issued currency could be 'kept stable by redeemability in a variable commodity reserve' (Hayek, 1943, p.176). Rather than pegging to a national currency, the issuer would hold and continually adjust a basket of goods so that the currency maintained constant purchasing power. This was more than a call for competition; it was a proposal for an independent monetary standard.

Historical evidence supports this insight. Studies of commodity and fiat regimes show that inflation is lower and more predictable under commodity-based standards. Rolnick and Weber (1997, p.1309) find that 'the average rate of inflation was lower under the commodity standard' and that price-level behaviour was more stable. White (2015, p.60) similarly reports that a commodity standard 'would deliver lower long-run inflation and greater price-level predictability than our present fiat standard'. These findings underscore that the crucial issue is not simply private issuance but the definition of the unit of account.

Austrian monetary theory makes the same point. Klein (1974, p.423) shows that private issuers can preserve purchasing power when redemption is tied to a real-value standard and reputation is at stake. Selgin (1988, pp. 4 – 6) demonstrates that historical free banking systems maintained stability through competitive issue when convertibility into real assets imposed discipline. Selgin and White (1994) add that 'the invisible hand' of market rivalry can indeed supply money when redemption is well defined.

Much of the stablecoin literature therefore narrows Hayek's argument. By relying on the 1970s pamphlets while neglecting the 1943 paper, it treats competition among dollar substitutes as sufficient and overlooks the deeper question of how to define money itself. The result is a system that is technologically innovative but conservative in monetary design. It gives users convenient blockchain access to fiat currency, but it does not advance the original promise of cryptocurrencies to create a genuinely non-state money.

A Hayekian flatcoin starts from his 1943 insight. It would be issued privately and competitively, but its value would be tied to a basket of goods selected to maintain stable purchasing power. Market prices, not central bank policy, would guide the continual adjustment of that basket. With modern blockchain

and oracle technology, this supply response can be automated, allowing Hayek's proposal to be implemented as a self-governing digital currency. The next section examines the original commodity reserve currency in detail and shows how it provides the foundation for an independent standard of value.

Hayek 1943: A Commodity Reserve Currency Revisited

In 1943 Hayek proposed what he called a 'commodity reserve currency' as a way of providing a money of constant purchasing power without the need for discretionary monetary management. Published in *The Economic Journal* and later reprinted in *Individualism and Economic Order*, the proposal was explicitly directed at the instability of both metallic and fiat standards. Hayek began from the observation that 'a currency redeemable on demand in a commodity unit of constant purchasing power is technically possible' (Hayek, 1943, p.176). His goal was to design a monetary standard in which the unit of account itself would embody the price signals of the market.

The mechanism was straightforward, but radical. The issuing institution would hold a reserve composed of many different commodities. The weights of these commodities would not be fixed but would vary with market conditions so that the average value of the basket, expressed in the chosen monetary unit, remained constant. Hayek explained that 'the composition of the commodity reserve would have to be continuously changed in such a way that its value remained stable' (Hayek, 1943, p.179). This was not a simple return to a gold standard, where a single commodity fixed the money supply. Nor was it a paper currency dependent on policy discretion. It was a rule-based system that used market prices to adjust the backing of the currency and to discipline its supply.

The proposal also recognised the importance of competitive issuance. Hayek envisaged that private banks would issue notes or deposits redeemable in the commodity unit and that competition among issuers would reinforce stability. He wrote that 'the issue of such a currency would best be left to private enterprise' and that 'competition between issuing institutions would provide an additional safeguard of stability' (Hayek, 1943, p.183). Convertibility into a well-defined basket of goods, combined with rivalrous issuance, would keep the quantity of money aligned with demand.

Hayek's article appeared alongside a short reply by John Maynard Keynes. In *The Objective of International Price Stability* Keynes endorsed the aim of a money of constant purchasing power but doubted whether a commodity-basket mechanism could deliver stability at the level of national prices. He argued that price levels are driven by domestic wage-cost conditions and that 'it is not possible to secure stability of the internal price level by fixing an external standard' (Keynes, 1943, pp.185–187). This exchange situates the commodity reserve currency within the wartime debate on international monetary reform and underscores the originality of Hayek's attempt to create a rule-bound, market-based standard at a time when most economists were focused on centralised planning of the post-war monetary order.

The commodity reserve currency was thus a concrete mechanism to achieve what Hayek (1945, p.521) elsewhere described as the key requirement of a monetary order; that money should transmit accurate information about relative scarcities and preserve the purchasing power of savings. It was also a response to the knowledge problem that Hayek had previously discussed. No central authority could know the correct quantity of money; only a system in which supply responded automatically to price signals could provide the necessary coordination.

Although outlined during wartime and before the development of electronic markets, Hayek's 1943 design anticipated later Austrian insights. Subsequent work on free banking and the competitive supply of money confirmed that private note issue tied to real assets could maintain convertibility and

stability without central direction (Klein, 1974; Selgin, 1988; Selgin and White, 1994). Historical studies likewise show that commodity standards produced more predictable outcomes than fiat regimes (Rolnick and Weber, 1997; White, 2015). These findings support Hayek's claim that a commodity-based standard could deliver a more reliable unit of value than any policy-managed fiat system.

The contrast with Hayek's 1970s writings is striking. *Choice in Currency* and *Denationalisation of Money* call for the abolition of the state monopoly of issue and for competition among private issuers, but they leave the monetary standard itself undefined (Hayek, 1976; 1976/1978). Nothing in those pamphlets prevents private issuers from offering liabilities denominated in state currencies. That is exactly what most stablecoins now do. The 1943 paper, by contrast, specifies a self-contained standard of value based on a commodity basket. It does not depend on the continued existence of national monies.

The commodity reserve currency was never adopted in Hayek's lifetime or since, largely because of the technical difficulty of maintaining and adjusting a diversified physical reserve. But the essential elements of the design – continuous market-based adjustment of a commodity basket, competitive private issuance, and redemption in a unit of stable purchasing power – are now technically feasible on a blockchain. Oracles can supply real-time price data, smart contracts can execute supply adjustments, and decentralised autonomous organisations can govern reserves and redemption rules without reliance on a central authority (Davidson, 2023; 2025a).

Recovering Hayek's 1943 proposal is therefore more than an exercise in history of thought. It provides the conceptual foundation for a truly non-state digital currency. Where present stablecoins merely compete in the issuance of dollar substitutes, a Hayekian flatcoin would compete in the standard of value itself. The next section develops this claim by translating the commodity reserve currency into a modern on-chain architecture and showing how the market signals that Hayek regarded as indispensable can be embedded directly in code.

Building a Hayekian Flatcoin

Hayek's 1943 proposal was limited by the technology of his time. Maintaining and adjusting a physical reserve of many commodities was costly and slow. Prices had to be collected from scattered markets and reserves physically stored and audited. Blockchain infrastructure now makes it possible to implement the same logic with lower cost and greater transparency.

A blockchain is a distributed ledger that records transactions in a sequence of cryptographically linked blocks. The ledger is maintained collectively by many independent nodes and is updated through a consensus protocol rather than a central authority. Every transaction is time-stamped, verified, and effectively irreversible. This architecture supplies a rule-bound environment in which property rights over digital tokens can be defined and enforced without a trusted intermediary. Hayek's vision of a monetary system governed by known rules and market signals rather than administrative discretion can therefore be instantiated in a blockchain.

Smart contracts are programmes that run on a blockchain and execute pre-specified actions when defined conditions are met. They can hold assets, enforce rules of issuance and redemption, and adjust the supply of tokens in response to external data. Decentralised autonomous organisations (DAOs) are arrangements of smart contracts that allow holders of a token to participate directly in governance. Rules for proposing changes, voting, and updating contracts are themselves written into the code. DAOs therefore create a transparent and enforceable structure for collective decision-making and can lower the cost of credible commitment (Davidson, 2023; 2025a; 2025b). Davidson, De Filippi and Potts (2018) describe blockchains as 'economic institutions of capitalism' that perform coordination and

enforcement roles traditionally reserved for firms, markets, or governments (p.641). A Hayekian flatcoin can draw on this institutional capacity to embed monetary rules directly in code.

Reliable price information is essential. Oracles are systems that bring off-chain data into a blockchain environment. They query external price feeds, aggregate the results, and deliver them to smart contracts. A flatcoin would rely on oracles to track the prices of commodities in its reserve basket. To protect against manipulation, the oracle system would need multiple independent data sources, cryptographic proofs of correctness, and a governance mechanism for resolving discrepancies.

A Hayekian flatcoin would be a digital token redeemable for a unit of stable purchasing power defined by a broad commodity basket. Its essential feature would be automatic adjustment of supply and reserves to keep the market value of the basket constant. Instead of physical warehouses, the reserve would be held as tokenised claims on commodities or on financial instruments that track commodity prices. Smart contracts would calculate the target index and expand or contract supply as needed. Holders could redeem their tokens for the equivalent value in the underlying commodity claims. This operational design gives effect to Hayek's insistence that the composition of the reserve must 'be continuously changed in such a way that its value remained stable' (Hayek, 1943, p.179).

Issuance and redemption would be competitive. As Hayek (1943) envisaged, private issuers would create and redeem tokens, with their reputation and capital at stake. Klein shows that in such an environment 'each issuer has an incentive to maintain the value of its currency at par with the standard' (Klein, 1974, p.423). Selgin (1988) likewise argues that free-banking systems maintain convertibility and price stability when competition disciplines issuers. In a blockchain setting, anyone could supply liquidity to redemption pools, and arbitrage would ensure that market prices remained close to the target index.

The question of adoption is not a mere technicality. Money must be accepted as a medium of exchange and store of value. The experience of Bitcoin shows that decentralised digital 'money' can emerge and gain value even without initial non-monetary uses. Hendrickson, Hogan and Luther (2016, p.926) describe how early adopters of Bitcoin coordinated on a common expectation of future value, allowing it to function as a medium of exchange. Luther provides further evidence that 'Bitcoin's early success can be explained as a coordination game in which entrepreneurs and users converged on a new monetary equilibrium' (Luther, 2019). These findings suggest that a Hayekian flatcoin could gain traction if it offered superior protection against inflation and clear redemption rights.

The macroeconomic implications of such a system are significant (Rolnick and Weber 1997; White 2015). A flatcoin that faithfully implements Hayek's commodity reserve logic could inherit the advantages of historical commodity-based regimes while using modern technology to overcome the operational constraints that limited those regimes.

A blockchain implementation can also address the problem of discretionary monetary policy that concerned Hayek in both his early and later work. By embedding the rules of supply adjustment in transparent smart contracts and making the reserves auditable on-chain, the system reduces the scope for manipulation. Governance research on DAOs shows how credible exit options and open participation can make capture by insiders more costly (Davidson, 2025a, p.3).

Designing such a system also requires attention to seigniorage and incentive structures. In traditional commodity standards, the cost of holding reserves limited excessive issue. In a flatcoin, smart contracts could require full collateralisation or set risk-based fees, ensuring that issuers internalise the cost of maintaining stability. The institutional challenge is to create a framework in which market participants supply stability as a by-product of profit seeking, not as a matter of administrative discretion.

A Hayekian flatcoin would therefore combine three elements. First, a clear and rule-bound definition of purchasing power based on a diversified commodity basket. Second, competitive issuance and redemption with strong incentives to maintain par value. Third, a decentralised governance structure that secures accurate price information and credible commitment to the rules. Each of these elements builds directly on Hayek's 1943 proposal, enriched by later Austrian monetary theory and made operational by blockchain technology.

Economic and Institutional Implications

A Hayekian flatcoin would not merely replicate the dollar on a blockchain. It would constitute a distinct monetary standard defined by a market-selected commodity basket and maintained through competitive private issuance. This prospect raises economic, institutional, and legal questions that differ from those posed by fiat-pegged stablecoins.

The economic dynamics of a flatcoin deserve careful consideration. Commodities can be volatile, so the currency's rule set must allow for continuous rebalancing and for temporary shocks that exceed normal adjustment capacity. Changes in the velocity of money could shift demand for liquidity and require the reserve mechanism to respond. Network effects also matter. Established fiat-based stablecoins enjoy deep trading pairs and user familiarity. A flatcoin will need to offer clear advantages, such as superior inflation protection and transparent governance, to overcome these incumbent advantages.

From a wider Austrian macroeconomic perspective, a Hayekian flatcoin would also help to prevent the intertemporal distortions that arise under fiat systems. Garrison (2000) shows how monetary manipulation by central banks misaligns interest rates with time preferences, distorting the capital structure and generating boom–bust cycles. Horwitz (2000) emphasises that because money enters into every exchange, monetary mismanagement creates economy-wide disequilibria rather than isolated market disturbances. By tying value to a commodity basket and removing discretion over supply, a flatcoin would provide a more reliable signal for saving and investment, reducing the risk of malinvestment and helping to coordinate economic activity across time.

An additional perspective is provided by other Austrian monetary theorists. Mises's regression theorem requires that the purchasing power of money be traceable to a commodity with non-monetary demand. A programmable basket of raw materials satisfies this condition because each component, be they metals, grains, or energy products, has established uses outside monetary exchange (Mises, 1949). Tokenisation does not sever the commodity connection; it merely changes the form of the claim, allowing smart contracts to make the underlying value transparent. In contrast to Bitcoin, whose initial value arose mainly from expectations of future exchangeability, a flatcoin would inherit immediate market value from the goods it represents.

Selgin's theory of free banking likewise fits the design – he shows how competitive note issuers maintain monetary equilibrium when liabilities are redeemable in an outside standard (Selgin, 1988). A programmable basket gives that external anchor, while smart contracts automate the redemption discipline and real-time supply adjustment that nineteenth-century clearinghouses enforced through custom and reputation. White's work reinforces the point. He documents how commodity standards historically delivered lower inflation and greater price-level predictability (White, 2015) and argues in *Better Money* (2023) that credible, objective monetary constitutions can emerge from market processes. A blockchain-based commodity basket, rule-bound and transparent, would be a modern example of such a constitution.

Taken together, these perspectives show that a Hayekian flatcoin does more than echo Hayek's own proposals. It realises a broader Austrian programme in which the value of money is grounded in goods with non-monetary demand and its supply is disciplined by competition and contract. Smart contracts and decentralised governance supply the technical means to implement what Mises, Selgin, and White describe in theoretical and historical terms.

The theory of competitive money supply helps explain how such a system can remain stable. Klein shows that when issuers compete and redeem liabilities in a non-state standard, each has an incentive to maintain par value to protect reputation and profits (Klein, 1974, p.423). Selgin demonstrates that free-banking systems can automatically match supply and demand without persistent inflation or deflation (Selgin, 1988, pp.4–6). Selgin and White describe how 'the invisible hand' of market rivalry can perform the monetary functions usually assigned to a central bank when convertibility is well defined (Selgin and White, 1994, p.1719). In a flatcoin, these mechanisms would operate through smart contracts and decentralised autonomous organisations rather than nineteenth-century clearinghouses.

Blockchain technology adds further institutional support. As Davidson notes, verifiable on-chain rules 'lower the cost of credible commitment' and allow governance to rely on market signals rather than administrative discretion (Davidson, 2023, p.144). A DAO structure, with transparent voting and credible exit options, can deter attempts to dilute collateral or manipulate price feeds (Davidson, 2025a). By embedding these safeguards in code, a flatcoin provides a more credible realisation of Hayek's requirement that the reserve 'be continuously changed in such a way that its value remained stable' (Hayek, 1943, p.179).

Legal and regulatory issues present another layer of complexity. A flatcoin would not fit neatly into existing categories of securities, commodities, or payment instruments. Historical evidence shows that the legal framework shapes monetary competition; Selgin documents how free banks maintained stability when private note issue operated within clear legal rights (Selgin, 1988), and Rolnick and Weber (1997) show how taxes and state-level restrictions affected bank behaviour. Modern regulators may also resist a currency that erodes state control over money. Allen, Berg and Davidson (2023) describe such reactions as 'repugnant innovation', where new practices meet hostility not only for technical reasons but because they challenge established norms and political authority (p.920). A Hayekian flatcoin, by depriving governments of their monetary monopoly, is a very likely candidate for such resistance.

The reaction to Facebook's Libra proposal underscores this point. Launched in 2019 and later rebranded as Diem, Libra was designed as a permissioned blockchain maintained by a Swiss association of corporate members. Every token was to be fully backed by a reserve of bank deposits and short-term government securities denominated in major fiat currencies, with issuance and redemption controlled to maintain par value (Amsden et al., 2019; Pierro and Tonelli, 2022; Jafari and Gruber, 2021; Simmons, 2021). Its stated purpose was to provide a global medium of exchange and low-cost cross-border payments to billions of social-media users (Pupolizio, 2021; Kianieff, 2021). Yet this very ambition provoked a coordinated international backlash. Central banks and finance ministries warned that a global private currency could undermine monetary sovereignty, destabilise financial markets, and frustrate anti-money-laundering enforcement (Pupolizio, 2021; Simmons, 2021). Under pressure, Libra scaled back to single-currency stablecoins, rebranded as Diem, and eventually abandoned its Swiss licence application before shutting down in 2022 (Jafari and Gruber, 2021; Pierro and Tonelli, 2022).

Libra/Diem illustrates the concept of repugnant innovation with unusual clarity. Although it introduced new technology and promised financial inclusion, it remained tied to the very fiat monies it was meant

to challenge. Its reserve basket consisted of government-issued currencies and debt, and its value was defined in those terms. Governance remained centralised in an association that could be targeted by regulators. Hayek's 1943 commodity-reserve currency, by contrast, envisaged a privately issued currency redeemable in a fixed basket of raw commodities, with automatic supply adjustment to keep real purchasing power constant (see figure 1 for a full comparison between the two proposals). Whereas Hayek aimed to sever the link between money and government finance, Libra's design preserved that link and so remained politically dependent (Kianieff, 2021; Schwarcz, 2022). The rapid suppression of the project confirms that any private currency anchored in state money is exposed to political veto and cannot escape discretionary control.

Figure 1: Comparison between Hayekian flatcoin and Libra

Feature	Hayek's 1943 proposal	Facebook's Libra/Diem
Reserve assets	Fixed basket of physical commodities to stabilise real purchasing power	Basket of fiat currencies and short-term government debt
Price anchor	Market prices of commodities, adjusting supply automatically	Exchange rates of state-issued currencies, requiring discretionary management
Governance	Competitive private issuers with no permanent central authority	Swiss-based association of corporate members with centralised decision-making
Relationship to state money	Explicitly severs the link between money and government finance	Remains defined in terms of state monies and dependent on banks and regulators
Exposure to regulation	Designed to operate outside political control	Directly subject to banking, securities, and payments law, making it vulnerable to regulatory veto

By creating a privately governed and rule-bound standard of value, a flatcoin would exert competitive pressure on official currencies. It would provide households and firms with a low-inflation medium of exchange and store of value, encouraging them to move savings and settlement away from national monies. Hayek anticipated such pressure when he argued that 'governments should be deprived of their monopoly of issuing money' (Hayek, 1976, p.22). The scale of adoption would depend on user confidence and on how effectively the system integrates with global payment networks.

The emergence of a Hayekian flatcoin would also enrich economic theory. It would illustrate Hayek's view that markets are discovery processes and that institutions evolve to make use of dispersed knowledge (Hayek, 1945). It would further confirm the argument that blockchain is an institutional technology capable of supplying missing pieces of economic infrastructure, a claim developed in recent research on decentralised autonomous organisations (Davidson, 2025c; 2025d).

A Hayekian flatcoin is therefore more than a technical proposal. It is an institutional alternative to the fiat monetary order, grounded in the historical experience of commodity standards and made possible by the programmable credibility of blockchain. By combining competitive issuance with a rule-based standard of value, it offers a way to achieve monetary stability without central banks. The concluding section considers how such a system could emerge and what it would mean for Austrian monetary theory and for the future of cryptocurrency.

Conclusion

This paper has argued that the prevailing model of stablecoins is a technological convenience but a monetary regression. By denominating liabilities in state fiat and relying on commercial banks and money-market instruments, they reproduce the discretionary inflation and institutional fragility of the current system. They are private payment rails for government money rather than a new monetary standard. Much of the cryptocurrency literature draws on Hayek's 1970s pamphlets, *Choice in Currency* and *Denationalisation of Money*, which emphasise competition in the supply of currency (Hayek, 1976; 1976/1978). Yet these writings leave the unit of account untouched and thereby allow the state's money to remain the anchor of value.

Hayek's earlier work points to a different path. In *A Commodity Reserve Currency* he described how a privately issued money could maintain constant purchasing power through continuous adjustment of a commodity basket (Hayek, 1943). This design anticipates the Austrian insight that monetary order must arise from dispersed knowledge and competitive discovery (Hayek, 1945). It also accords with the historical record showing that commodity standards deliver lower and more predictable inflation than fiat standards (Rolnick and Weber, 1997; White, 2015). The 1943 exchange with Keynes confirms that the proposal was recognised in its own time as a serious alternative for achieving price stability (Keynes, 1943).

Modern blockchain infrastructure now makes it possible to implement Hayek's proposal in a way that was not feasible in 1943. Decentralised oracles provide real-time price data. Smart contracts can execute the rules of supply and redemption. Governance by decentralised autonomous organisations offers credible commitment and resistance to capture (Davidson, 2023; 2025a). Bitcoin's history shows that decentralised digital money can gain acceptance without state backing (Hendrickson, Hogan and Luther, 2016; Luther, 2019). Together these developments create the conditions for a Hayekian flatcoin: a currency defined by a commodity-based unit of value and maintained by competitive private issuance on a blockchain.

Important economic and institutional challenges remain. Commodity prices can be volatile and may at times move faster than any automatic rebalancing rule can accommodate. Network effects give incumbent stablecoins deep liquidity and user familiarity that a flatcoin must overcome. Changes in monetary velocity may create unexpected pressures on supply. Legal and regulatory barriers are likely: a flatcoin would not fit neatly into categories of securities, commodities, or payments instruments, and it would represent what Allen, Berg and Davidson (2023, p.920) describe as 'repugnant innovation,' where a novel practice provokes political or social resistance precisely because it undermines entrenched authority. These are engineering and institutional problems to be solved, not reasons to abandon the design.

A Hayekian flatcoin would not simply add another trading pair to digital markets. It would create a new monetary standard. By embedding the definition of value and the rules of supply in transparent code, it would limit discretionary inflation and dispense with central bank management. It would advance Hayek's aim of 'depriving governments of their monopoly of issuing money' (Hayek, 1976, p.22) and would move closer to Satoshi Nakamoto's ambition to create electronic cash that does not rely on financial institutions (Nakamoto, 2008).

This proposal is not a call to return to gold. It shows how a neglected idea in Hayek's monetary theory can be combined with the institutional possibilities of blockchain to create a currency of stable purchasing power. By recovering the logic of the commodity reserve currency and applying it to a decentralised architecture, a Hayekian flatcoin offers a practical and theoretically grounded route to a non-state digital money. For Austrian economics it illustrates how entrepreneurial innovation can extend the market order. For cryptoeconomics it provides a path beyond fiat replication toward a genuine monetary alternative.

Appendix: Glossary of Blockchain Terms

The argument of this paper makes use of concepts drawn from blockchain engineering and cryptoeconomics that may be unfamiliar to some readers. To avoid interrupting the flow of the main text, this appendix provides a short glossary. Each term is explained in intuitive rather than technical language, with an emphasis on the economic functions these mechanisms perform.

Term	Intuitive Explanation
Blockchain	A distributed record-keeping system maintained by many independent computers rather than a single authority. Each new block of records is cryptographically linked to the previous one, making the history tamper-evident. Economically, it functions as a ledger that no single party can unilaterally alter.
Distributed ledger	Another way of describing blockchain. It emphasises that the record is replicated across many computers, making the system resilient to failure or tampering at any single point.
Consensus protocol	The rules by which blockchain participants agree on which transactions are valid and in what order they occurred. Protocols such as Proof-of-Work and Proof-of-Stake allow decentralised agreement without relying on a central authority.
Smart contract	A computer program stored on a blockchain that automatically executes actions when pre-set conditions are met. Unlike traditional contracts that depend on courts or intermediaries, smart contracts enforce themselves in code. Example: "If the commodity index rises above X, release payment Y."
Decentralised Autonomous Organisation (DAO)	An organisation whose governance rules are encoded in smart contracts rather than set by executives or boards. Token holders can propose and vote on changes, and approved decisions execute automatically. DAOs replace managerial discretion with transparent, programmable rules.
Oracle	A mechanism for bringing external information (such as commodity prices, interest rates, or weather data) onto a blockchain. Oracles are necessary because blockchains cannot access off-chain data directly. Their credibility depends on minimising manipulation and aggregating multiple sources.
Token	A unit of value issued on a blockchain. Tokens can represent money, commodity claims, voting rights, or access permissions. They are transferable between users without a central registry or clearinghouse.

Tokenisation	Converting ownership rights to real-world assets (such as commodities, real estate, or securities) into blockchain tokens. The underlying asset remains the same; what changes is that claims are represented digitally and can be exchanged securely on a blockchain.
On-chain / Off-chain	On-chain refers to data or activity recorded directly on a blockchain, making it transparent and verifiable. Off-chain refers to information or activity outside the blockchain, such as dollars in bank accounts or physical commodities in warehouses.
Collateral (crypto context)	Assets locked in a smart contract to back the value of issued tokens. Unlike traditional collateral held by an institution, blockchain collateral is managed by code and auditable in real time. If requirements are not met, the smart contract can liquidate collateral automatically.
Redemption	The process of exchanging a token for the underlying asset or value it represents. In a flatcoin system, redemption means converting tokens into commodity claims. For credibility, redemption must be automatic and enforceable without depending on an issuer's discretion.
Liquidity pool	A pool of tokens locked in a smart contract to facilitate exchange and redemption. Users contribute assets to the pool and receive fees in return, ensuring continuous trading without central intermediaries.
Stablecoin	A blockchain-based token designed to maintain a stable value, usually pegged to a state-issued currency like the US dollar. Stability depends on the reserves and custodians backing it.
Flatcoin	A proposed alternative to stablecoins. Rather than being pegged to fiat money, a flatcoin would be tied to a diversified commodity basket, with blockchain-based rules and governance ensuring stability of purchasing power.

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